ENGINEERING MINI-PROJECT REPORT

Water Quality Prediction Using Machine Learning

Classification-based water potability prediction
using physical and chemical water-quality parameters

Submitted by

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PROJECT TYPE	Classification
PRIMARY MODEL	Logistic Regression
TARGET	Potability (0 = Not Potable, 1 = Potable)

Executive Summary

Safe drinking water is an important engineering and public-health requirement. Water quality depends on multiple physical and chemical characteristics, and a sample may be difficult to classify using a single parameter. A data-driven classification system can provide a preliminary screening mechanism by learning patterns from historical water-quality measurements.

This project develops a supervised machine-learning classification workflow for predicting whether a water sample is potable or not potable. The dataset contains 3,276 water-quality observations with nine physicochemical input variables and one binary target variable. The inputs are pH, hardness, dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity.

The implementation performs dataset inspection, exploratory analysis, missing-value treatment, feature preparation, standardization, an 80:20 stratified train-test split, and Logistic Regression training. Because the target classes are imbalanced, the final Logistic Regression model uses balanced class weights. The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Final reported test metrics: Accuracy = 52.44%, Precision = 41.46%, Recall = 53.13%, and F1-score = 46.58%. The results indicate that the model is able to identify both classes after balancing, but its predictive performance is not strong enough for independent water-safety certification.

Important engineering interpretation: the model should be considered a preliminary decision-support or screening tool. A prediction of potable water should not replace laboratory testing, regulatory testing, or certified water-treatment monitoring.

1. Introduction

Drinking-water quality is affected by acidity or alkalinity, dissolved minerals, disinfectant-related chemicals, organic matter, conductivity, and suspended or dissolved constituents. Monitoring these parameters is therefore an engineering task involving measurement, analysis, and decision-making.

Machine learning provides a way to combine several measurements and learn a relationship between water-quality parameters and an observed potability label. In this project, the objective is deliberately limited to binary classification rather than claiming a complete water-treatment or certification system.

The project follows the engineering workflow specified for the mini-project: problem definition, data understanding, data preparation, ML technique selection, model development, evaluation, engineering interpretation, and final recommendation.

2. Problem Definition

The engineering problem is to classify a water sample as potable or not potable using measured physicochemical parameters. The intended system accepts numerical measurements from a water sample and produces a binary classification.

The problem is formulated as classification because the target variable, **Potability**, contains two classes: 0 for not potable and 1 for potable.

2.1 Project Domain

Water-quality monitoring, environmental engineering, drinking-water screening, and applied machine learning.

2.2 Objectives

- Understand the structure and quality of the water-potability dataset.
- Inspect missing values, duplicates, descriptive statistics, class distribution, and relationships among variables.
- Prepare the numerical measurements for supervised classification.
- Use the nine physicochemical parameters as model inputs.
- Train a simple and interpretable Logistic Regression classifier.
- Address the observed class imbalance through balanced class weights.
- Evaluate the model using accuracy, precision, recall, F1-score, and confusion matrix.
- Interpret the model result in the context of preliminary water-quality screening.
- Identify limitations and practical improvements for a more reliable deployment.

2.3 Expected Inputs

pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity.

2.4 Expected Output

Binary water-potability classification: **Potable** or **Not Potable**, together with the model's class probabilities for a new sample.

2.5 Engineering Decision Supported

The prediction can support preliminary screening and help identify samples that may require additional inspection or laboratory testing. It is not intended to certify water as safe for human consumption.

Why this matters: water-safety decisions have direct engineering and public-health consequences. Therefore, a machine-learning prediction should be interpreted as decision support and validated against accepted laboratory and regulatory procedures.

3. Dataset Description

The project uses the commonly distributed Water Potability dataset containing 3,276 observations and 10 columns: nine numerical physicochemical features and one binary target. The dataset is associated with the Kaggle Water Potability dataset by Aditya Kadiwal.

Dataset Property	Value
File	water_potability.csv
Records	3,276
Original columns	10
Predictor variables	9
Target variable	Potability
Not Potable (0)	1,998 (60.99%)
Potable (1)	1,278 (39.01%)
Missing data	Present in pH, Sulfate and Trihalomethanes
Duplicate rows	0

3.1 Dataset Variables

Feature	Description	Role
ph	Acidity/alkalinity of water.	Predictor
Hardness	Dissolved calcium and magnesium related hardness.	Predictor
Solids	Total dissolved solids in the water.	Predictor
Chloramines	Chloramine concentration associated with water treatment.	Predictor
Sulfate	Dissolved sulfate concentration.	Predictor
Conductivity	Electrical conductivity related to dissolved ions.	Predictor
Organic_carbon	Organic carbon content.	Predictor
Trihalomethanes	Disinfection by-products measured in the sample.	Predictor
Turbidity	Measure related to water clarity.	Predictor
Potability	Binary indicator of whether the sample is potable.	Target

4. Data Exploration

The dataset was inspected before model development to understand its dimensions, variable types, missing values, class distribution, and numerical ranges. Exploratory analysis is important because preprocessing and model choices should be justified by the observed data rather than applied blindly.

4.1 Exploratory Analysis Performed

- Dataset shape and structure inspection.
- Data-type and column inspection.
- Descriptive statistics for numerical variables.
- Missing-value analysis by column.
- Duplicate-row check.
- Potability class distribution analysis.
- Distribution and correlation analysis.

4.2 Key Dataset Observations

Variable	Mean	Minimum	Maximum
pH	7.0808	0.0000	14.0000
Hardness	196.3695	47.4320	323.1240
Solids	22014.0925	320.9426	61227.1960
Chloramines	7.1223	0.3520	13.1270
Sulfate	333.7758	129.0000	481.0306
Conductivity	426.2051	181.4838	753.3426
Organic Carbon	14.2850	2.2000	28.3000
Trihalomethanes	66.3963	0.7380	124.0000
Turbidity	3.9668	1.4500	6.7390

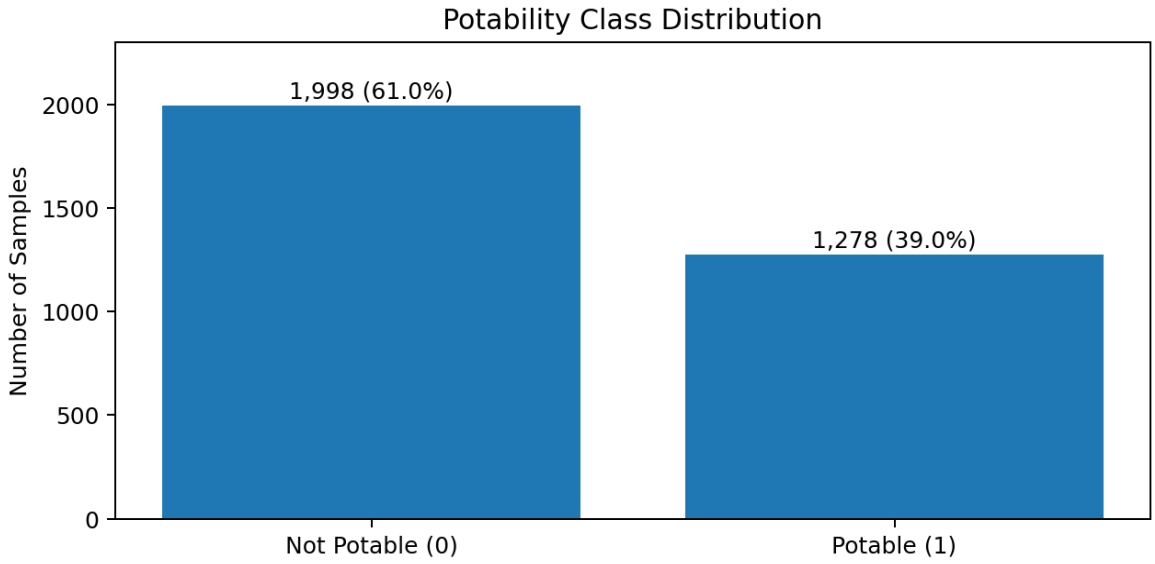


Figure 1. Distribution of potable and non-potable observations.

5. Data Preprocessing

Preprocessing is required because several predictor variables contain missing observations and because Logistic Regression benefits from standardized numerical features. The preprocessing decisions are kept simple and directly connected to the observed dataset.

5.1 Missing-Value Treatment

Column	Missing Values	Treatment
ph	491	Median imputation
Sulfate	781	Median imputation
Trihalomethanes	162	Median imputation
Other predictors	0	No imputation required
Potability	0	No imputation required

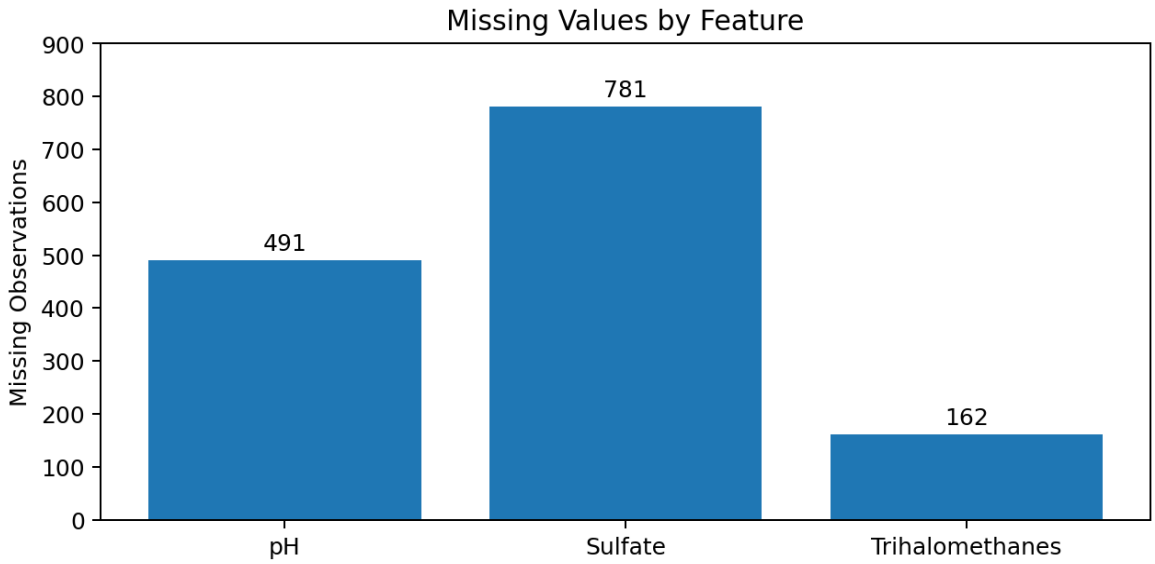


Figure 2. Missing observations identified before imputation.

Median imputation is used because it provides a simple robust replacement for missing numerical values and is less sensitive to extreme observations than the arithmetic mean.

5.2 Duplicate Removal

The dataset contains no duplicate rows in the inspected version, so no duplicate observations were removed.

5.3 Train-Test Separation

The data is divided using an 80:20 stratified train-test split with random_state = 42. The stratification preserves the relative proportion of potable and non-potable samples in both subsets. This produces approximately 2,620 training observations and 656 test observations.

Why this matters: the test set remains unseen during model training and provides a consistent basis for evaluating how the trained classifier behaves on new observations.

5.4 Feature Scaling

StandardScaler is fitted using the training data and then applied to both training and test data. Scaling prevents features with larger numerical magnitudes, such as Solids, from dominating the Logistic Regression optimization merely because of their units.

6. Feature Engineering and Feature Selection

The project intentionally uses a compact and physically interpretable feature set. No complex derived variables are introduced. All nine measured physicochemical variables are retained because each represents a different aspect of water quality relevant to preliminary potability assessment.

Predictor	Reason for Inclusion
ph	Represents acidity/alkalinity and chemical condition of the water.
Hardness	Represents dissolved mineral characteristics.
Solids	Represents total dissolved material in the sample.
Chloramines	Provides information about disinfectant-related chemistry.
Sulfate	Represents dissolved sulfate concentration.
Conductivity	Provides an electrical indication associated with dissolved ions.
Organic_carbon	Represents organic matter content.
Trihalomethanes	Represents disinfection by-products.
Turbidity	Represents water clarity and suspended-material effects.

7. Machine Learning Technique

Binary classification is the appropriate primary formulation because the target variable contains two classes. Logistic Regression is selected as the project model because it is simple, fast, interpretable, and appropriate for a beginner-level engineering classification task.

7.1 Algorithm Selection

Model	Purpose	Strength	Limitation
Logistic Regression	Binary classification	Simple, fast, interpretable probability-based classifier	May not capture complex nonlinear relationships.

7.2 Logistic Regression Pipeline

Median Imputation → StandardScaler → Logistic Regression

The scaler and imputer are fitted using training data so that information from the test set does not influence the preprocessing parameters.

7.3 Class Balancing

The target distribution is approximately 61% not potable and 39% potable. The final model therefore uses **class_weight='balanced'**. This increases the relative importance of the minority class during training and reduces the tendency to predict only the majority class.

> **Why this matters: the initial unbalanced Logistic Regression behavior was strongly biased toward the majority class. Balanced class weights produced a model that identifies both classes, although the overall accuracy remained modest.**

8. Implementation

The project is implemented in Python using a Jupyter Notebook or Google Colab environment. The implementation uses standard data-science and machine-learning libraries and avoids unnecessary advanced methods.

Component	Technology / File
Programming language	Python
Notebook	main.ipynb / Google Colab notebook
Dataset	water_potability.csv
Data handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine learning	Scikit-learn
Final model	Logistic Regression with balanced class weights
Report	Water_Quality_Prediction_Engineering_Mini_Project_Report.pdf

8.1 Processing Workflow

- Load the water-potability dataset.
- Inspect dimensions, data types, descriptive statistics, missing values, and duplicates.
- Explore the target-class distribution and numerical relationships.
- Separate the nine predictors from the Potability target.
- Split the data into training and test subsets using an 80:20 stratified split.
- Fit median imputation using the training data.
- Standardize the predictor variables using the training data.
- Train Logistic Regression with balanced class weights.
- Generate test-set predictions and class probabilities.
- Calculate accuracy, precision, recall, and F1-score.
- Interpret the confusion matrix and engineering implications.
- Use the trained model to classify a new water sample.

8.2 New Sample Prediction Capability

The completed notebook also contains a function that accepts the nine water-quality measurements for a new sample, applies the same imputation and scaling transformations, and returns the predicted class together with the probabilities of the two classes.

9. Results

The final Logistic Regression model was evaluated on the held-out test set. The model uses balanced class weights because the original dataset contains more non-potable than potable observations.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression (balanced)	52.44%	41.46%	53.13%	46.58%

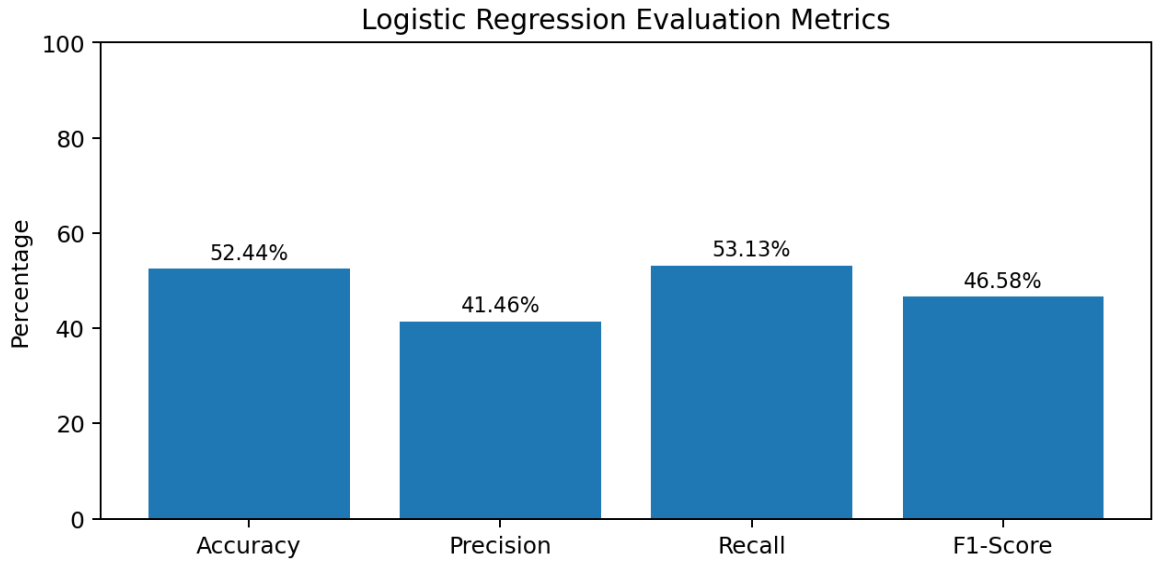


Figure 3. Final Logistic Regression evaluation metrics.

9.1 Result Interpretation

- The model achieves 52.44% accuracy on the held-out test data.
- Precision of 41.46% indicates that a substantial proportion of samples predicted as potable are not actually potable.
- Recall of 53.13% indicates that the model identifies slightly more than half of the potable samples in the test set.
- The F1-score of 46.58% shows a moderate balance between precision and recall, but the overall predictive performance remains limited.
- The balanced model is preferable to the initial unbalanced behavior because it predicts both target classes rather than collapsing toward the majority class.

The results should therefore be presented as a baseline classification outcome rather than as evidence of a production-ready water-safety system.

10. Evaluation

Evaluation is performed using multiple classification metrics because accuracy alone can be misleading when the target classes are imbalanced. Precision, recall, and F1-score provide additional information about how the classifier behaves for the potable class.

10.1 Evaluation Metrics

Metric	Meaning in this project
Accuracy	Percentage of all test samples classified correctly.
Precision	Among samples predicted as potable, the proportion that are actually potable.
Recall	Among actually potable samples, the proportion correctly identified by the model.
F1-Score	Harmonic balance between precision and recall.

10.2 Evaluation Strategy

The model is trained only on the training subset and evaluated on a held-out stratified test subset. The test data is not used to fit the imputer, scaler, or Logistic Regression coefficients. This separation provides a more realistic estimate of performance on unseen samples.

10.3 Confusion Matrix

The notebook includes a confusion matrix to visualize true-positive, true-negative, false-positive, and false-negative predictions. For water-quality screening, false-positive predictions are particularly important because classifying unsafe water as potable could create a misleading safety signal.

	Predicted Not Potable	Predicted Potable
Actual Not Potable	True Negative	False Positive
Actual Potable	False Negative	True Positive

Note: The exact cell counts should be copied from the executed notebook's final confusion-matrix output before submission if the department requires the numerical matrix in the printed report.

11. Engineering Interpretation

The practical value of this project is not limited to the numerical classification metrics. The model combines multiple water-quality measurements to provide a preliminary estimate of potability. This can be useful for screening and prioritizing samples for further testing.

- A predicted 'Potable' result indicates that the measured combination of parameters resembles patterns associated with potable samples in the training data.
- A predicted 'Not Potable' result can be used as a trigger for additional inspection, treatment review, or laboratory testing.
- Because precision is only 41.46%, potable predictions from this model should not be interpreted as certification of safety.
- Recall of 53.13% indicates that the model still misses a substantial fraction of potable samples, so the model is not suitable as the sole screening criterion.
- The class-balanced training strategy improves attention to the minority class but does not by itself guarantee high predictive performance.
- The model can support an engineering workflow in which ML screening is followed by laboratory verification and regulatory decision-making.

11.1 New Water Sample Prediction

The notebook supports a new-sample prediction workflow. A sample can be entered using pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. The system applies the same preprocessing steps used during training and returns a predicted class and class probabilities.

Example Input	Value
pH	7.0
Hardness	200
Solids	20,000
Chloramines	7
Sulfate	300
Conductivity	400
Organic Carbon	14
Trihalomethanes	70
Turbidity	4

Engineering caution: the output of this example is a machine-learning prediction, not a laboratory water-quality result.

12. Limitations

- The final Logistic Regression model has limited predictive performance, with 52.44% accuracy and 46.58% F1-score.
- The dataset is imbalanced, with approximately 61% non-potable and 39% potable observations.
- Missing values in pH, sulfate, and trihalomethanes require imputation, which may introduce uncertainty.
- The dataset does not contain sufficient geographic, temporal, source-water, or treatment-process context to establish broad real-world generalization.
- The binary Potability label does not describe the degree or type of contamination.
- The model is not validated against a laboratory certification process and should not be used as a replacement for regulatory water testing.
- The current project uses a single baseline classification algorithm and does not include hyperparameter tuning or cross-validation.

13. Recommendations and Future Improvements

- Collect larger and more geographically diverse water-quality datasets.
- Include source location, sampling time, treatment-stage information, and laboratory reference measurements.
- Compare Logistic Regression with approved nonlinear classifiers if model comparison is permitted.
- Use stratified cross-validation to obtain a more robust estimate of generalization performance.
- Tune model parameters only after establishing a strong baseline.
- Investigate class-specific thresholds when the cost of false-positive or false-negative decisions is known.
- Add input validation to detect physically impossible or clearly corrupted measurements.
- Integrate the trained model into a simple dashboard or web application for controlled demonstration.
- Validate any deployed model against current laboratory measurements before operational use.

14. Conclusion

This engineering mini-project developed a complete machine-learning workflow for water-potability prediction. The project began with an engineering problem definition and dataset inspection, followed by exploratory analysis, missing-value treatment, feature preparation, scaling, stratified train-test separation, and Logistic Regression model development.

The final balanced Logistic Regression model achieved 52.44% accuracy, 41.46% precision, 53.13% recall, and 46.58% F1-score on the held-out test set. The model therefore provides a useful educational baseline and demonstrates the complete supervised-classification workflow, but its current performance is not sufficient for independent water-safety decisions.

From an engineering perspective, the main value of the system is its ability to combine multiple water-quality measurements into a preliminary screening prediction. Any real deployment should use the model as a supporting component within a validated water-quality monitoring process and should retain laboratory and regulatory verification as the final authority.

References

1. Kadiwal, A. Water Potability Dataset, Kaggle. Dataset used for the project: water_potability.csv.

2. Pune Vidyarthi Griha's College of Engineering, Technology and Management, Department of Electronics and Telecommunication Engineering. Experiment 10 – Mini Project guidelines, Academic Year 2026–27.

3. Scikit-learn documentation, LogisticRegression, StandardScaler, train_test_split, classification metrics.

Appendix A – Requirement Compliance Checklist

Requirement	Status	Evidence
Engineering problem defined	Completed	Section 2
Dataset documented	Completed	Section 3
Data quality investigated	Completed	Sections 4 and 5
At least 3 meaningful visualizations	Completed	Figures 1–3
Preprocessing decisions justified	Completed	Section 5
Feature preparation explained	Completed	Section 6
ML algorithm justified	Completed	Section 7
Model trained and evaluated	Completed	Sections 8–10
Engineering interpretation	Completed	Section 11
Limitations identified	Completed	Section 12
Recommendations provided	Completed	Section 13
Final conclusion	Completed	Section 14

Appendix B – Final Submission Checklist

- Attach the executed main notebook with visible outputs.
- Attach or reference the approved water-potability dataset.
- Verify that all numerical metrics in the report exactly match the final notebook.
- Insert the final confusion-matrix image from the executed notebook if numerical matrix values are required.

- Ensure the report, notebook, README, and project folder use the same model and preprocessing workflow.
- Confirm dataset approval requirements before final departmental submission.

FINAL PROJECT SUMMARY

Project: Water Quality Prediction Using Machine Learning

Item	Final Project Decision
Problem Type	Binary Classification
Dataset	Water Potability Dataset – 3,276 samples
Features	9 physicochemical parameters
Target	Potability (0 / 1)
Preprocessing	Median imputation + StandardScaler
Train/Test	80:20 stratified split
Model	Logistic Regression, class_weight='balanced'
Accuracy	52.44%
Precision	41.46%
Recall	53.13%
F1-Score	46.58%
Engineering Use	Preliminary screening / decision support
Deployment Caution	Laboratory and regulatory verification remains necessary

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